Jian Han

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Research Interests

AI-enabled power-system analysis
Large language model agents, automated modeling workflows, and decision-support tools for grid studies.

- Power-system automation
- Engineering workflow agents
- Interpretable analysis tools

Stability, optimization, and resilience
Methods for voltage stability assessment, optimization-based analysis, and resilient operation of modern power networks.

- PV and QV curve analysis
- Weak-grid diagnosis
- Resilience-oriented planning and operation

Experience

State Grid Corporation of China, Power Systems Engineering Experience China
Worked close to practical grid operation and planning problems, motivating research on usable analytical tools for power-system engineers.

- Power-system operation and planning context
- Grid-analysis workflows
- Practical engineering constraints

Projects

LLM for Automated Modeling and Analysis in Power Systems
A research direction exploring how large language model agents can assist with power-system modeling, study setup, result interpretation, and reproducible analysis.

- Automated study preparation
- Model-aware reasoning
- Engineer-facing explanations

PV Curve Agent
An interactive assistant for voltage stability studies, PV curve interpretation, and weak-bus or weak-area diagnosis.

- Voltage stability assessment
- PV curve workflow support
- Transparent analysis outputs

Publications

AI-Enabled Voltage Stability Assessment and Power-System Analysis
Details to be updated as the manuscript and conference metadata become public.
Jian Han

Skills

Power Systems

Optimization and AI

Research Software